

File Permissions in Linux

Project Description

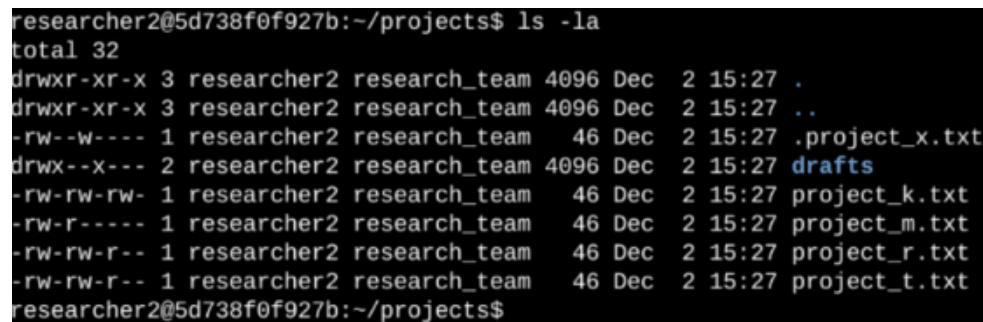
In this activity, I worked with Linux file and directory permissions for a research team. The goal was to check the existing permissions and make sure users only had the access they were supposed to have. I used Linux commands to check the permissions, identify unnecessary access, and update the permissions where needed.

Check File and Directory Details

I used the `ls -la` command to check the permissions of the files and directories inside the projects directory.

```
ls -la
```

Screenshot 1 — Command and output:



```
researcher2@5d738f0f927b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 ..
-rw--w---- 1 researcher2 research_team  46 Dec  2 15:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Dec  2 15:27 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Dec  2 15:27 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Dec  2 15:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_t.txt
researcher2@5d738f0f927b:~/projects$
```

The `ls -la` command displays the detailed information for all files and directories, including hidden files. The `-l` option shows the detailed permission information, while the `-a` option includes hidden files such as `.project_x.txt`.

The first column in the output contains a 10-character permission string. This string shows the file type and the permissions given to the user, group, and others.

Describe the Permissions String

The 10-character permission string can be divided into different parts:

- **1st character:** Shows the file type. `d` means it is a directory, while `-` means it is a regular file.
- **2nd–4th characters:** Show the permissions for the user/owner: read (`r`), write (`w`), and execute (`x`).
- **5th–7th characters:** Show the permissions for the group.
- **8th–10th characters:** Show the permissions for others, meaning users who are not the owner or part of the group.
- `A -` means that particular permission is not allowed.

For example:

```
-rw-rw-r--
```

The first character `-` shows that this is a regular file. The owner has read and write permissions, the group also has read and write permissions, and others only have read permission. No one has execute permission on this file.

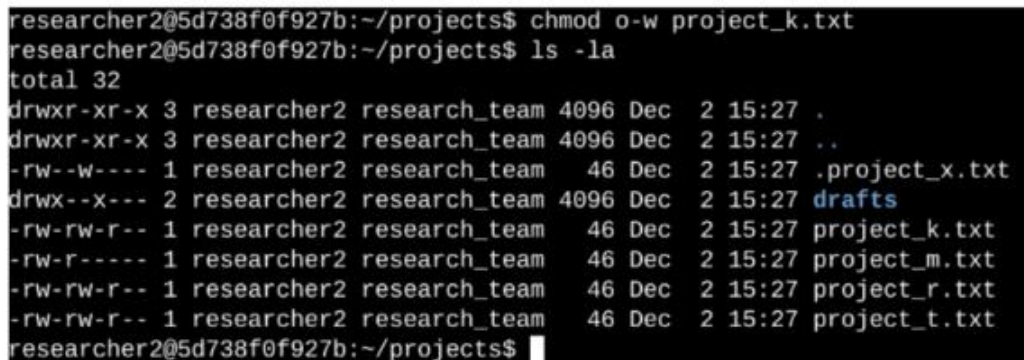
Change File Permissions

The organization does not allow other users to have write access to files. After checking the existing permissions, I found that `project_k.txt` had write permission for others.

I used the following command to remove that permission:

```
chmod o-w project_k.txt
ls -la
```

Screenshot 2 — Removing write permission from others:



```
researcher2@5d738f0f927b:~/projects$ chmod o-w project_k.txt
researcher2@5d738f0f927b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 ..
-rw--w---- 1 researcher2 research_team  46 Dec  2 15:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Dec  2 15:27 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Dec  2 15:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_t.txt
researcher2@5d738f0f927b:~/projects$
```

The `chmod` command is used to change permissions. In this command, `o` represents others and `-w` removes write permission. I then used `ls -la` again to confirm that the permission had been updated.

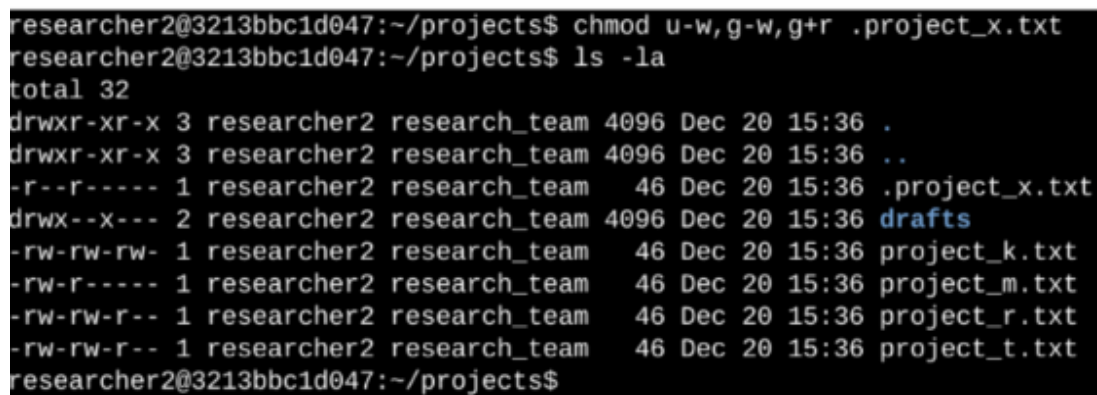
Change File Permissions on a Hidden File

The research team had archived `.project_x.txt`. Because the filename starts with a period (`.`), it is a hidden file. The requirement was that the user and group should be able to read the file, but nobody should have write permission.

I used the following commands to update its permissions:

```
chmod u-w .project_x.txt
chmod g-w .project_x.txt
chmod g+r .project_x.txt
ls -la
```

Screenshot 3 — Updating permissions for the hidden file:



```
researcher2@3213bbc1d047:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@3213bbc1d047:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec 20 15:36 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec 20 15:36 ..
-r--r----- 1 researcher2 research_team  46 Dec 20 15:36 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Dec 20 15:36 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Dec 20 15:36 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Dec 20 15:36 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec 20 15:36 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec 20 15:36 project_t.txt
researcher2@3213bbc1d047:~/projects$
```

The `u-w` option removes write permission from the user, while `g-w` removes write permission from the group. I used `g+r` to make sure the group had read permission. Finally, I used `ls -la` to check the updated permissions, including the hidden file.

Change Directory Permissions

Only the `researcher2` user should be able to access the `drafts` directory and its contents. Other users should not have execute permission on this directory.

I used the following command:

```
chmod g-x drafts
ls -la
```

Screenshot 4 — Updating the drafts directory permissions:

```
researcher2@5d738f0f927b:~/projects$ chmod g-x drafts
researcher2@5d738f0f927b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 ..
-r--r----- 1 researcher2 research_team  46 Dec  2 15:27 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Dec  2 15:27 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Dec  2 15:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_t.txt
researcher2@5d738f0f927b:~/projects$
```

The `g-x` option removes execute permission from the group. The `researcher2` user already had execute permission, so I did not need to add it. After making the change, I used `ls -la` to verify the updated permissions.

Summary

I first used `ls -la` to examine the existing permissions, including hidden files. I then used `chmod` to remove unnecessary write and execute permissions from files and directories. This helped make the permissions match the required level of access for the research team and reduced unnecessary access to the project files.